

# A survey of numerical algorithms for trajectory optimization of flight vehicles

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As one of the important components of computational flight mechanics and control, numerical algorithms of trajectory optimization for flight vehicles are currently studied by many researchers in aerospace engineering to completely solve these difficult problems, but few papers on the survey of this research field have been published recently. Based on the investigation of more than one hundred literatures, considering the application perspectives of computational flight mechanics and recent developments of trajectory optimization, the numerical algorithms of trajectory optimizations for aerospace vehicles are summarized and systematically analyzed. This paper summarized the basic principle, characteristics and application for all kinds of current trajectory optimization algorithms; and introduced some new methods and theories appearing in recent years. Finally, collaborative trajectory optimization for many flight vehicles, and hypersonic vehicle trajectory optimization were mainly reviewed in this paper. In the conclusion of this paper, the future research properties are presented regarding to numerical algorithms of trajectory optimization and control for flight vehicles as follows: collaboration and antagonization for many flight vehicles and multiple targets, global, real-time online, high accuracy of 7-D trajectory, considering all kinds of unknown random disturbances in trajectory optimization, and so on.

**computational flight dynamics, flight trajectory optimization, numerical algorithms, survey, global optimization**

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## 1 Introduction

Flight vehicles are the instruments flying in atmosphere or (and) space outside atmosphere. Flight vehicles flying outside atmosphere mainly include satellites, space-based spacecraft systems, moon exploration spacecrafts and deep space exploration systems. Aircrafts inside atmosphere mainly include space-launched vehicles, (civil, military) aircrafts, all kinds of missiles (ground-to-ground, land-to-air, ground-to-air, air-to-ground and various cruising), unmanned aircraft, intermediate altitude (near space) spacecraft and microminiature aeronautical vehicles. Flight vehi-

cle trajectory optimization is decided by many factors, including the kinds and characteristics of flight vehicles, flight mission (the objective performances) and neighboring environment of flight vehicles. The flight missions or performance indexes mainly include the collaboration and (/or) antagonization of numerous flight vehicles, minimum fuel consumption, minimum take-off weight, maximum or minimum flight time, passing moving (or/and) static target points, avoiding dynamic (or/and) static threat areas, rendezvousing with some target points (in the same space point with same velocity), minimum requirements for flight vehicle characteristics (flight vehicle characteristics include available overload factor, lift-to-drag ratio, thermal protection system, engine propulsive properties, computer inside vehicle and communication system). Flight neighboring

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environment refers to various gravitational fields, atmosphere characteristics (wind field, atmospheric pressure and temperature field), function of wireless and electromagnetic field, other possible relevant flight vehicles and various random disturbances.

The difficulties in flight vehicle trajectory optimization are numerous flying collaboration and antagonization, the multi-objectives and global optimization of flight trajectories (and static parameters) for numerous/single flight vehicles, real-time online, high accuracy, and considering various unknown random disturbances.

The global and integrative optimization of flight trajectory and static design parameters is important content of modern flight vehicle design, which can be in favor of improving flight qualities to meet the technical requirements for all determined missions. With the expansion of flight vehicle application field, the higher accuracy of trajectory optimization has to be obtained in design process, especially in the design of spacecraft and high-altitude, super-speed and highly mobile flight vehicles. Trajectory optimization plays an important role in the design process of whole flight vehicle, having an influence on the design of various sub-systems, including overall layout, aerodynamic configuration, guidance control, motive power and structure [1]. The key factor affecting the success or failure of flight vehicle system design is to dynamically and statically assemble and match all sub-systems of flight vehicle, to best achieve performance index, such as minimum take-off weight, maximum payload, lowest cost or maximum reliability, which is of great significance and practical value in flight vehicle design [2].

Trajectory optimization is to form flight mechanical equation based on specified tactical technology index and to choose main design parameter to construct performance functional to get optimal parameter by using relevant mathematical methods to form optimal flying trajectory. In the mathematics essence, space trajectory optimization can be abstracted to be the optimal control problem of solving the open cycle of functional extreme value under the restraint of differential equation, algebraic equation and inequality. Thus, the generation of the optimal space trajectory is the problem of solving relevant optimal control [1].

The theory of space trajectory optimization has developed from the 1950s to 1970s. The main mission is to resolve the approximate analytic solution or suboptimal numerical solution. With the rapid development of optimal control theory, optimal numerical computation solution and computer technique, the numerical calculation of optimal trajectory has developed significantly in the 1980s, and was driven to a maturity stage. In general, the available numerical calculations of space trajectory optimization include gradient method, conjugate gradient method, generalized reduced gradient method, projection recovery gradient method, advanced conjugate gradient method, climbing method, variable metric method, direct substitution method,

variational method, differential dynamic method, nonlinear programming, multiple shooting method, simplex method and an optimal control method of using parameter optimization [2]. The classification of trajectory optimization methods is different because of its standards. Several classification standards are [3] direct method and indirect method, integration method and differentiation method, single-objective optimization and multiple-objective optimization, guaranteeing optimization and robust optimization, and pure mathematics method and impure mathematics method.

Summarizing literature about trajectory optimization method is a rare thing. A widely quoted paper is the one Betts published in 1998 [4]. He summarized the primitive trajectory optimization numerical methods, including direct method and indirect method. Papers published by Yong et al. [3] and others in 2008 summarized fundamental algorithm, characteristics and application of trajectory optimization numerical methods. The latest development in this field introduced some new fundamental theories and characteristics of space trajectory optimization. In the paper published in 2009, Chen et al. [1] systematically summarized and analyzed direct numerical calculations which are presently used in space trajectory optimization, discussed classification of direct method as two steps (discrete transformation and nonlinear programming), and introduced the fundamental theory, advantages and disadvantages of direct shooting method, collocation method, differential inclusion method. Finally the paper resumed the application of direct method.

By the view of the application of flight mechanics and on the basis of recent research achievement in trajectory optimization, Section 2 of this paper describes the problem of trajectory optimization for multiple flight vehicles. Section 3 summarizes fundamental principles, characteristics and applications of trajectory optimization methods and introduces new methods and theories in recent years. Regarding to the current popular and sensitive problems about the collaborative path planning of multiple flight vehicles, and trajectory optimization constrained by non kinetic variables (such as aerodynamic heating and energy) for hypersonic vehicles, Sections 4 and 5 specifically summarize the trajectory optimization of hypersonic vehicle and the collaborative trajectory planning for numerous flight vehicles. Section 6 summarizes the whole paper and comes up with research direction for trajectory numerical optimization of flight vehicles in future.

## 2 Problem description

There are many problems in flight vehicle trajectory optimization, including various flight vehicles and trajectory optimizations, various shuttle missions (performance index, optimization objective), complex and uncertain flying sur-

rounding environment and complex and changeable constraint conditions. Various trajectory optimizations of flight vehicles can be reduced to that a large stochastic flight control system of various flight vehicles contains  $S$  flight vehicle subsystems. The whole stochastic flight control system of various flight vehicles can be described by the following nonlinear simultaneous differential equations:

$$\begin{cases} \frac{d\mathbf{x}_1(t)}{dt} = \mathbf{f}_1[t, \mathbf{X}(t), \mathbf{u}_1(t), \mathbf{w}_1(t), \mathbf{p}_1], \\ \frac{d\mathbf{x}_2(t)}{dt} = \mathbf{f}_2[t, \mathbf{X}(t), \mathbf{u}_2(t), \mathbf{w}_2(t), \mathbf{p}_2], \\ \dots\dots\dots \\ \frac{d\mathbf{x}_S(t)}{dt} = \mathbf{f}_S[t, \mathbf{X}(t), \mathbf{u}_S(t), \mathbf{w}_S(t), \mathbf{p}_S], \end{cases} \quad (1)$$

where  $\mathbf{X}=[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_S]$ , in which, the time  $t \in [t_{i0}, t_{if}]$ ,  $i=1, 2, \dots, S$ , the state variable  $\mathbf{x}_i(t)$  is the  $n_i$  dimensional vector of the  $i$ th flight vehicle (i. e. subsystem  $i$ ,  $\mathbf{x}_i(t) \in R^{n_i}$ ), the state variable  $\mathbf{u}_i(t) \in R^{m_i}$  is the independence variable of the  $i$ th flight vehicle, the random disturbance  $\mathbf{w}_i(t) \in R^{L_i}$  impacts the subsystem  $i$ ,  $\mathbf{p}_i \in R^{Q_i}$  is the static parameters of the flight vehicle  $i$  (not changing with the time).  $\mathbf{f}_i(\dots)$  is the motion function of each flight vehicle system considering dynamic random subsystem presenting the mutual interference.

The initial condition of each flight vehicle system  $i$  at the time  $t_{i0}$  (from starting point O) is as follows:

$$\mathbf{x}_i(t_{i0}) = \mathbf{x}_{i0}, \quad \mathbf{u}_i(t_{i0}) = \mathbf{u}_{i0} \quad (2)$$

(at the time  $t_{if}$ ), the flight vehicle system  $i$  moves to the terminal point F, and the following terminal constraints have to be satisfied:

$$\mathbf{x}_i(t_{if}) = \mathbf{x}_{if}, \text{ or } \begin{cases} N_1[\mathbf{x}_i(t_{if}), t_{if}] = 0, \\ N_2[\mathbf{x}_i(t_{if}), t_{if}] \leq 0, \end{cases} \quad (3)$$

at the same time the flight process will be restrained by the following process conditions:

$$\mathbf{g}_{i,\min} \leq \mathbf{g}_i(t, \mathbf{X}, \mathbf{u}_i, \mathbf{W}) \leq \mathbf{g}_{i,\max}, \quad (4)$$

where  $\mathbf{g}_i$  is the  $n_{g_i}$  dimensional vector function,  $\mathbf{W}=[\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_S]$ ,  $\mathbf{u}_i(t)$  and  $\mathbf{a}_i(t)$  are respectively restrained by  $\Omega_U$  and  $\Omega_p$ , which are represented as follows:

$$\mathbf{u}_{i,\min} \leq \mathbf{u}_i(t) \leq \mathbf{u}_{i,\max}, \quad (5)$$

$$\mathbf{p}_{i,\min} \leq \mathbf{p}_i \leq \mathbf{p}_{i,\max}. \quad (6)$$

The random disturbance  $\mathbf{w}_i(t)$  in each flight vehicle sub-

system  $i$  takes an evolutionary motion according to the following differential equation:

$$\frac{d\mathbf{w}_i(t)}{dt} = \mathbf{h}[t, \mathbf{W}(t), \mathbf{v}_i(t), \mathbf{X}(t)], \quad (7)$$

where  $i=1, 2, \dots, S$ , the starting point of  $\mathbf{w}_i(t)$  is  $\mathbf{w}_i(t_{i0}) = \mathbf{w}_{i0}$ ,  $\mathbf{v}_i(t)$  is the control variable of the random disturbance  $\mathbf{w}_i(t)$ , it can be random, and also be designed.

In the motion process, there are  $T$  performance indexes for each flight vehicle as follows:

$$J_{i,j}[\mathbf{u}_i(t)] = \int_{t_{i0}}^{t_{if}} L_{i,j}[t, \mathbf{X}(t), \mathbf{u}_i(t), \mathbf{w}_i(t), \mathbf{p}_i] dt, \quad (8)$$

where  $i=1, 2, \dots, S$ ,  $j=1, 2, \dots, T$ ,  $[t_{i0}, t_{if}]$  is the working period of the  $i$ th flight vehicle, the terminal constraint (also represented as performance index) of which is

$$J_{i,T+1}[\mathbf{u}_i(t)] = \Phi_i[\mathbf{x}_i(t_{if}), t_{if}], \quad (9)$$

where  $\Phi_i[\mathbf{x}_i(t_{if}), t_{if}]$  is the cost function at the terminal point.

The optimal control problem refers to searching the input control function  $\mathbf{U}^*(t)=[\mathbf{u}_1^*(t), \mathbf{u}_2^*(t), \dots, \mathbf{u}_S^*(t)]$  and the static parameters  $\mathbf{p}_i$ , and handles the random subsystems (1) of the  $S$  flight vehicles affected by random disturbance (7) at the time  $[t_{i0}, t_{if}]$ , the large scale random control system moves from the initial starting point condition (2) to the terminal condition (3), and is impacted by eqs. (4)–(6), each flight vehicle system moves along with the trajectory  $\mathbf{X}^*(t)=[\mathbf{x}_1^*(t), \mathbf{x}_2^*(t), \dots, \mathbf{x}_S^*(t)]$ , which makes all of the following multiple performance indexes reach global optimal (the minimum):

$$\min_{\mathbf{U}(t) \in \Omega_U} J = \min \begin{bmatrix} J_1 \\ \mathbf{u}_1(t) \\ J_2 \\ \mathbf{u}_2(t) \\ \dots \\ J_S \\ \mathbf{u}_S(t) \end{bmatrix} = \min \begin{bmatrix} J_{1,1}, & J_{1,2}, & \dots & J_{1,T+1} \\ \mathbf{u}_1(t) & \mathbf{u}_1(t) & & \mathbf{u}_1(t) \\ J_{2,1}, & J_{2,2}, & \dots & J_{2,T+1} \\ \mathbf{u}_2(t) & \mathbf{u}_2(t) & & \mathbf{u}_2(t) \\ \dots & \dots & \dots & \dots \\ J_{S,1}, & J_{S,2}, & \dots & J_{S,T+1} \\ \mathbf{u}_S(t) & \mathbf{u}_S(t) & & \mathbf{u}_S(t) \end{bmatrix}, \quad (10)$$

where  $\mathbf{U}(t)=[\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_S(t)]$ . It is worthy of being pointed out that, if one of the performance indexes  $J_{i,j}$  ( $i \in [1, S]$ ,  $j \in [1, T+1]$ ) in function (10) needs to be maximized, the maximized  $J_{i,j}$  can be turned into the minimized

$$J_{i,j} = -J_{i,j}. \quad (11)$$

Therefore, in function (10), the  $S \times (T+1)$  performance indexes  $J_{i,j}$  are to mix up the problems of achieving maximization and minimization.

### 3 Numerical methods review

#### 3.1 Direct method

From the 1950s, Понтрягин and Bellman's outstanding contribution to optimal control theory laid a solid foundation for trajectory optimization research [1]. At the same time, with the stimulation of requirements from industrial application field, Bryson [5] began his research of optimal control theory. When minimum value principle is used to resolve optimal control problem, differential-algebraic equations transformed from optimal control problem are usually the two-point boundary value problem of nonlinear differential equations, the resolution of which is extremely difficult. Even for the indirect shooting method, near extreme value method and other numerical methods have been developed after the 1980s, the problems still exist, such as high sensitivity for original valuation and solution difficulty for complex constraint conditions [6]. Therefore, the way of resolving optimal trajectory by direct iteration and based on optimal control theory earns researchers' attentions.

Direct method includes two problems: how to transform optimal control problem to nonlinear programming problem, and how to resolve transformed nonlinear programming problem. The two problems are respectively discrete transformation and optimized resolution [1].

Direct method does not need the necessary condition to resolve optimal solution. It is to disperse and parameterize continuous optimal control problem and directly use numerical methods to optimize performance index [3]. Compared with indirect method, direct method is more widely applied and is of more transformation methods to realize parameterization foresaid. The main current algorithms for trajectory optimization include direct shooting method, multiple direct shooting method, point collocation method and differential inclusion method [4].

##### 3.1.1 Direct shooting method

Direct shooting method is of simple and clear thinking, which can directly use differential equations to conduct resolution. However, there are many existing shortcomings, such as sensitivity to initial value and large gradient computation. Direct method is suitable for simple optimal control problem of low requirement for accuracy. Traditional direct shooting method can only solve initial value problem [7].

The most prominent characteristics of direct shooting method are to disperse control variable in time domain and obtain state variables with numerical integration to resolve objective functions, and then transform optimal control problem to nonlinear programming problem [1].

As for optimal control, choose the limited time points  $T = \{t_1, t_2, \dots, t_k, \dots, t_N\}$ , where,  $k = 1, 2, \dots, N$ ,  $t_k \in [t_0, t_f]$ .

Let the value of predictive control variable  $\tilde{u}(t)$  at discrete point be  $\tilde{u}_{t_k} = \{\tilde{u}_{t_0}, \tilde{u}_{t_1}, \dots, \tilde{u}_{t_k}, \dots\}$ . Control value between discrete time-points is probably needed in numerical

integration. Thus, control variable needs to be interpolated. For example, the most common interpolation method is the linear algorithm.

$$\tilde{u} = \tilde{u}_{t_k} + \frac{\tilde{u}_{t_{k+1}} - \tilde{u}_{t_k}}{t_{k+1} - t_k}(t - t_k), \quad t_k \leq t \leq t_{k+1}. \quad (12)$$

Let control variables be

$$X = [\tilde{u}_{t_0}^T, \tilde{u}_{t_1}^T, \dots, \tilde{u}_{t_N}^T]^T. \quad (13)$$

With numerical solution of state differential equations in  $[t_0, t_f]$ , we obtain

$$\tilde{x}_i = F(X), \quad \tilde{J} = J(\tilde{x}, \tilde{u}, t) = J(X), \quad (14)$$

where  $\tilde{x}_i$  is the discrete series obtained from numerical solution.

The improvement for direct shooting method includes: choosing more accurate interpolation method (thrice interpolation, spline interpolation), choosing numerical integral method of higher accuracy to resolve differential equation, choosing nonlinear programming (NLP) calculation of higher accuracy and better convergence, and choosing numerical differential calculation of better performance.

The optimized variables of the NLP model transformed from direct method are of small amount. However, the form of objective function is undefined. When gradient needs to be calculated using the NLP calculation, a large amount of numerical gradient calculation is needed. With the increase of the amount of discrete points, the calculation amount is increasing. At the same time, it is hard to prove that the NLP solution from common optimal control problem can be converged to system optimal control.

Direct shooting method has obtained good results in solving rocket ascent trajectory and transfer orbit optimization of flight vehicle.

##### 3.1.2 Multiple direct shooting method

The numerical integrating range in direct shooting method is simple domain  $[t_0, t_f]$ . When the time interval is comparatively long, integral accuracy is usually lower. At the same time, numerical differential calculation is of worse effect. Multiple direct method adopting multi-segment integral technique is an advanced method of direct shooting method [8, 9].

The characteristic of this method is to disperse control variable and state variable, according to the chosen  $m$  points  $T' = \{t'_1, t'_2, \dots, t'_j, \dots, t'_m\} \subset T$ ,  $j = 1, 2, \dots, m$ .  $T$  is the discrete time-point concourse for the selected control variable foresaid.

When the numerical integration comes to  $t'_j \in T'$ , it begins a new integration. With this method, the error between state value and predictive value obtained in the state discrete joint during NLP optimization can be eliminated. Thus, the

calculation accuracy is improved.

### 3.1.3 Collocation method

Collocation method is the method of dispersing control variable and state variable in  $T$  at the same time, and transforming optimal control problem to NLP. There are many discrete methods in collocation method. As an example, this paper takes the arithmetic used in optimal trajectories by implicit simulation (OTIS). For simplicity, state differential equation is scalar equation for short. At the same time, state and control variables are scalar for short.

Set the control variables and state variables in discrete point respectively as

$$\begin{cases} \tilde{u}_{i_k} = \{\tilde{u}_{i_0}, \tilde{u}_{i_1}, \dots, \tilde{u}_{i_k}, \dots\}, \\ \tilde{x}_{i_k} = \{\tilde{x}_{i_1}, \tilde{x}_{i_2}, \dots, \tilde{x}_{i_k}, \dots\}. \end{cases} \quad (15)$$

Set optimal control variables as

$$X = [\tilde{u}_{i_0}^T, \tilde{u}_{i_1}^T, \dots, \tilde{u}_{i_N}^T, \tilde{x}_{i_1}^T, \dots, \tilde{x}_{i_N}^T]^T. \quad (16)$$

Set  $\tilde{x}_{i_k}$  to be an independent variable, apply numerical integral method in every domain among discrete points. State and control variable values in the needed region can be obtained by thrice polynomial interpolation. The value of objective function can be obtained by the superposition of piecewise integration and applying constraint conditions to discrete points.

$$\int_{t_k}^{t_{k+1}} f dt = \frac{t_{k+1} - t_k}{6} \left( f(t_k) + 4f\left(\frac{t_{k+1} + t_k}{2}\right) + f(t_{k+1}) \right). \quad (17)$$

At the same time, state differential equation can be transferred into implicit form.

$$g(X) = \int_{t_0}^{t_f} f(x, u, t) dt - x - x_{i_0} = 0, \quad (18)$$

where  $g(X)$  in NLP model is disposed as constraint condition.

Obviously, optimal variable dimension of NLP model obtained by collocation method outclasses that by direct shooting method. However, at the cost of which, morbidity degree of objective function is greatly decreased, the convergence and accuracy are improved.

The discrete value of state variable is taken as unknown variable. Thus, the process of collocating numerical integration is a series of symbolic operation, but is not completely resolved by numerical routine. The amount of symbolic operation needed in the transformation of a complex optimal control model is very astonishing, which is extremely obvious when there are many discrete points. The initial software OTIS integrated a series of NLP models transformed from vehicle flight trajectory optimization problems. But OTIS is not suitable because of other optimal control problems, which makes the application of collocation

method limited. The gradual maturity of Computer Algebra System (CAS for short, similar to Maple's computer assistant symbol arithmetic system) improved this condition. Based on MATLAB of symbolic operation and numerical analysis, a batch of optimal control solver softwares has appeared, the collocation method has been widely applied.

At present, pseudospectral method (PS) is a branch rapidly developed in collocation method. Its main characteristic is that the main part of discrete points concourse is Gauss point (for example, Legendre PS method adds initial point and terminal point based on Gauss points concourse. Radau PS method adds initial point based on Gauss points set). Orthogonal polynomial is used to approximate control variable and state variable. PS method is of low sensibility of initial value to numerical iterative. And it is of good convergence.

Under the certain conditions, applying PS method to transform optimal control problem meets the so-called Costate Mapping Principle (CMP for short), which means that under those conditions, the results of direct method and indirect numerical method are unified. Thus, the convergence of PS method is ensured by the academic view.

Whether approximation polynomial and the selection of collocation points meet CMP of control variable and state variable in PS method is described in Table 1 [10].

### 3.1.4 Differential inclusion

Differential inclusion is a discretization method based on implicit integration theory, which can be taken as the improvement of collocation method. Compared with collocation method, variable in NLP model transformed from differential inclusion is less. In general, the calculation speed is much faster by using differential inclusion, which is more beneficial to the online and real time implementation of trajectory optimizations for flight vehicles. Thus, differential inclusion is a discrete transformation method with broad prospect [10].

The prominent characteristic of differential inclusion is that it only discretizes state variables. Constrained control variables can be eliminated by constraining the change rates of state variables. But control variables must be offset in differential inclusion. Thus, it is difficult to transform complex system, which makes it difficult to be widely applied in parameterization [4].

Direct method has the following advantages:

(1) It is not necessary to deduct the first rank of optimality

**Table 1** Contrast of different PS methods

| PS method   | Discrete point type | Interpolation polynomial | Meeting CMP or not |
|-------------|---------------------|--------------------------|--------------------|
| LPM         | LGL point           | Lagrange polynomial      | no                 |
| GPM         | LG point            | Lagrange polynomial      | yes                |
| CPM         | GGL point           | Chebyshev polynomial     | no                 |
| RPM         | LGR point           | Lagrange polynomial      | no                 |
| Hermite-LPM | LGL point           | Lagrange polynomial      | no                 |

condition.

(2) Compared with indirect method, direct method has broader convergence domain and lower requirements for the accuracy of initial value evaluation. Guessing of the initial value of consistent variable is unnecessary. And switching of structure prior information is unnecessary.

Disadvantages of direct methods are:

(1) Many direct methods do not provide master vector information. Thus, the obtained nonlinear programming solution cannot be ensured to be the solution of the original optimal control problem.

(2) The direct method of control variable parameterization is easy to converge local optimal solution, which depends on the initial guess.

Taking overall optimal capability of genetic algorithm into consideration, some scholars use genetic algorithm in solving NLP trajectory optimization. But genetic algorithm has the problem of large amount of calculation. In addition, as for hypersonic vehicle, its flight trajectory is of high sensibility to control parameter. Thus, trajectory optimization based on NLP method is of great difficulty. Compared with methods of only limiting control variable parameterization, point collocation is less sensible to initially predicted value [11].

Over decades, the application of direct numerical algorithm of trajectory optimization does a great favor in the development of aerospace industry. Reentry vehicle trajectory optimization has been solved by using internal optimization methods, such as continuous systematic sampling discrete method, sequential quadratic programming (SQP) and interior point method [12, 13].

Program to optimize simulated trajectories (POST) has solved ascent trajectory optimization and mission analysis of many launched vehicles (such as "Titan", "Delta" launcher) by using direct shooting method. With the application of direct methods, many problems have been solved, including the ascent, gliding and cruising trajectory optimization of hypersonic vehicle [14, 15], ascent trajectory optimization constrained by RLV [16], Sönger trajectory optimization [17] and trajectory optimization of constant thrust for the Moon landing [18].

Boeing Company uses Cheby Top procedure based on collocation method to optimize the trajectory of many satellites changing orbit of low thrust. Collocation method also solves many problems, including trajectory optimization of commercial plane, trajectory optimization of lunar soft landing and reentry trajectory optimization of spacecraft [1]. From the 1980s to 1990s, Boeing Company developed OTIS and SOCS softwares to apply collocation method in various flight vehicle trajectory designs.

On November 5 in 2006, the International Space Station conducted a large scale attitude adjustment. This is different from the adjustment before, with the mission of "zero-propellant engine driven" based on PS optimal control theory; this adjustment conducted previous trajectory optimi-

zation, which is the first large-scale application of PS method. PS method can not only save propellant cost of 1.5 million dollars for NASA, but also can realize the mission, which cannot be accomplished by the existing control calculation of the International Space Station [19].

So far, PS method has been widely applied to solving many optimal control problems, including trajectory programming of unmanned plane, missile guidance, trajectory optimization of mechanical arm, inverted pendulum stability and swaying, orbital maneuver of spacecraft and ascent stage guidance of injection spacecraft, and it becomes the most effective instrument to solve optimal control and trajectory optimization [20].

The development of direct method is closely connected with aerospace and industrial application. The research development of direct method theory can rapidly stimulate practical projects.

### 3.2 Indirect method

Indirect method transforms optimal control problem to a Hamiltonian boundary value problem based on Pontryagin minimum modulus principle. Thus, first of all, let us introduce Hamilton function [10]:

$$H[x(t), u(t), \lambda(t), t] = L[x(t), u(t), t] + \lambda^T(t) f[x(t), u(t), t], \quad (19)$$

where  $\lambda^T(t) = [\lambda_1(t), \lambda_2(t), \dots, \lambda_n(t)]$  is called adjoint vector or coordination vector. In order to be suitable for minimum modulus principle, boundary conditions can be disintegrated to be initial condition  $x(t_0) = x_0$  and terminal constraint  $N[x(t_f), t_f] = 0$ . If the necessary condition of optimal control  $u^*$  is the existence of non-zero vector function  $\lambda^*(t)$ ,  $t \in [t_0, t_f]$ . Let  $u^*(t)$ ,  $\lambda^*(t)$  and  $x^*(t)$ ,  $t_f^*$  meet the following requirements:

(1) Hamilton equations

$$\begin{cases} \dot{x}^* = \frac{\partial H[x^*(t), u^*(t), \lambda^*(t), t]}{\partial \lambda} \\ \dot{\lambda}^* = -\frac{\partial H[x^*(t), u^*(t), \lambda^*(t), t]}{\partial x} \end{cases} \quad (20)$$

(2) Minimal value condition

$$H[x^*(t), u^*(t), \lambda^*(t), t] = \min_{u \in U} H[x^*(t), u(t), \lambda^*(t), t]. \quad (21)$$

(3) Terminal transversality condition

$$\begin{cases} x(t_0) = x_0, \\ \lambda(t_f) = \left[ \frac{\partial \Phi}{\partial x} + \left( \frac{\partial N_1^T}{\partial x} \right) v \right]_{t=t_f}, \\ N_1[x(t_f), t_f] = 0. \end{cases} \quad (22)$$

(4) When terminal time  $t_f$  is changeable, terminal transversality condition used to determine  $t_f$  is

$$\left[ H + \frac{\partial \Phi}{\partial t} + \nu^T \frac{\partial N_1}{\partial t} \right]_{t=t_f} = 0. \quad (23)$$

Firstly, the expression of optimal control variable should be obtained in indirect method, which is the function related with adjoint variable and state variable. And then, two-point boundary value problem composed by Hamilton equations, terminal transversality condition and constraints should be resolved to obtain the numerical solution of optimal trajectory and relevant optimal control.

The difference between initial value and two-point boundary value is that all the initial values of differential equation state variable of the former are known. Numerical integration method is used to integrate differential function to obtain arithmetic solution. As for two-point boundary value problem, in  $2n$  simultaneous equations, there are only  $n$  known initial values of differential equation. The other  $n$  initial values of differential equations are unknown. And  $n$  terminal conditions are known. In order to conduct numerical integration according to the demand of the principle of initial value problem,  $n$  variable initial values should be selected randomly, thus  $2n$  differential equations can be conducted numerical integration to obtain solution at the same time. It is very difficult to resolve two-point boundary value problem. With the exception of some simple problems, accurate solution is hard to obtain. Numerical integration method can only be used to get approximate solution, because the initial value chosen randomly is usually hard to meet the other boundary condition. Moreover, the nonlinear system of ordinary differential equations of flight vehicle movement is complex. When flight vehicle is flying in atmosphere, the solution of trajectory optimization and guidance law is extremely complex because of the serious nonlinearity of aerodynamic force and atmosphere parameter, a large amount of table functions, various types of constraint conditions caused by control variables and state variables. Under such kind of condition, it is much harder to resolve accurate solution. At present, two numerical methods widely used in the resolution of two-point boundary value problem are shooting method and finite difference method [21].

### 3.2.1 Shooting method

In the solution of two-point boundary value problem, initial value of adjoint differential equations is given by the predicted value. Then, forward integration method is used to solve initial value. If the solution obtained can meet all the boundary conditions of the other terminal point, it is the solution of this boundary problem. Otherwise, new initially predicted value should be chosen to conduct new resolution. This solution process of the repeated iterative operation is terminated, only when the solution of two-point boundary value problem can meet all the boundary conditions. During

the usage of shooting method, it is extremely important to choose proper strategy and make full use of results of previous steps to adjust new initial value, which makes iterative solution converge rapidly to optimal solution. In general, there are many different kinds of available strategies and calculations. But the most common calculation and strategy is Newton-Raphson method. In the process of resolving two-point boundary value problem, Jacobian matrix needs to be calculated. Factor in matrix is not in the form of analytic function. However, Jacobian matrix of analytic expression is not needed in Newton-Raphson method. Thus, it is suitable for two-point boundary value solution.

### 3.2.2 Finite difference method

Flight vehicle trajectory optimization and guidance law are a group of nonlinear ordinary differential equations with variable coefficients. Thus, it can be approximately replaced by grid point value in integrating range by using difference equation. The trial solution is composed of dependent variable of each grid point. Although the selected grid point does not always meet finite difference equation, any value in grid point can be adjusted to get closer and closer. Finally, it can meet the solution of finite difference equation and boundary condition. It is remarkable that using finite difference method to solve two-point boundary value problem is related with grid point distribution, the form of using difference equation to express differential equation and iteration strategy. In general, under the condition of relatively smooth and nonviolent oscillation, infinite difference method is better than shooting method. Otherwise, shooting method can be used.

In the recent years, it is difficult to resolve two-point boundary value problem by using shooting method. The research of indirect method began to pay attention to how to effectively resolve two-point boundary value problem, such as the research of initially predicted technology of consistent variable [22], using genetic algorithm to resolve two-point boundary value problem [23] and advanced neighboring extreme method [24].

Advantages of indirect method are: (1) High accuracy of solution; (2) the optimal solution meets first-order optimal necessary condition.

Disadvantages of indirect method are: (1) It is complex to deduce the optimal solution based on maximum (minimum) value theory; (2) The convergence domain is extremely small during two-point boundary value problem resolving. Thus, high accuracy is required in initial value estimation of unknown boundary condition. And many indirect methods are required to guess the initial value of consistent variable. However, those variables haveno physical significance, which increases difficulty in initial value estimation; (3) As for optimal control problem with path constraint, it is difficult to use indirect method. The available solving way is to transform process constraint to equivalent terminal constraint or prior information or switch fabric of

constraint and non-constraint segment by mechanical translation.

Maximum theory is widely applied in previous trajectory optimization research. In the 1970s, Vinh [25] began to do research in optimal flight trajectory in atmosphere and problems based on variation principle and Pontryagin extremum principle, such as optimal unpowered flight, supersonic cruise, supersonic turn and optimal gliding of reentry spacecraft. Many problems take plane motion model into consideration, and reentry problem applied the amended Chapman formula. Istratie [26] applied indirect method to do research in different optimal objectives, such as minimum heating-flow density, maximum residual velocity and other optimal reentry trajectory problems. Vinh and Lu [27] deducted the method to resolve Chebyshev problem based on maximum value theory. They transformed problems to constraint problem of state variables along trajectory. Wu and Wang did a farther research on optimal air powered orbital transfer based on maximum value theory. And their research results are systematically introduced in ref. [28]. Relevant applications of indirect method also include optimal mechanical trajectory in airfight [29], trajectory optimization of hypersonic velocity gliding flight vehicle [23] and ascent trajectory optimization of launched vehicle.

When indirect method is used to resolve optimal control problem, Pontryagin maximum value theory should be used to deduct the first-order necessary condition of optimal control to obtain HBVP of resolving optimal trajectory. Then, direct shooting method and point collocation method mentioned in Section 3.1 are used to parameterize HBVP. Indirect method is of high solution accuracy and the optimal resolution meets the first-order optimal necessary condition. Compared with direct method, indirect method has some drawbacks in optimal control problem resolving: complex process of deducting optimal solution and transversality condition based on Pontryagin maximum value theory. The convergence domain is extremely small and high accuracy is required in initial value estimation. And state variable has no practical physical significance. Thus, it is difficult to predict initial value. As for optimal control problem includ-

ing trajectory constraint, trajectory constraint must be transformed to be the terminal constraint. When there is implicit constraint condition, the equivalent Lagrangian multipliers are hard to be eliminated, which will largely increase calculation quantity.

Based on systematical research and summary on the theory and results of trajectory optimization numerical calculation, Table 2 gives a comparative analysis on various direct and indirect methods mentioned above [10].

### 3.3 Dynamic programming

Dynamic programming is a mathematical method to resolve multistage decision process optimization. In the 1950s, based on a kind of multistage decision, Bellman et al. [30, 31] came up with the idea of taking optimal theory as the theoretical foundation of dynamic programming. The optimal theory is: the sub-strategy of an optimal strategy is always optimal. Its mathematical description is

$$\begin{cases} \min_{u_1} \min_{u_2} \cdots \min_{u_N} J(u_1, u_2, \cdots, u_N), \\ s.t. \dot{x} = f(x, u, t), \\ x(t_0) = x_0. \end{cases} \quad (24)$$

The advantage of dynamic programming is: simple calculation theory, relatively high accuracy and strict theory support.

Betts in his review quoted a description from ref. [4] to explain the disadvantage of this method. The largest drawback of this method is the curse of dimensionality just as Bellman himself said. Even for a relatively complex problem, record of the calculated results needs a large amount of memory space. If we just want to find optimal trajectory from a certain initial point, it is a cumbersome process to find overall initial value point with dynamic programming method. In addition, the selection of discrete form has no unified standard. The degree of discretization influences computation speed and accuracy.

Dynamic programming is one of the methods early used

**Table 2** Comparative analysis on various numerical optimal methods

| Method                 | Initial value sensitivity | Symbol deduction type                                                                                                        | NLP model        | Whether $t_f$ or not, uncertain problem | Accuracy        | Whether BVP or not |
|------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|-----------------|--------------------|
| Direct shooting        | high                      | simple discrete control variable                                                                                             | relatively small | relatively hard                         | low             | cannot             |
| Point collocation      | relatively low            | large calculation quantity of discrete control variable and state variable, but programming is possible.                     | extremely large  | relatively easy                         | relatively high | can                |
| Differential inclusion | relatively low            | hodograph mapping and discrete state variable, artificially deducting hodograph mapping and state discrete can be programmed | large            | relatively easy                         | relatively high | can                |
| Indirect               | relatively high           | first-order optimal condition, hard to be programmed                                                                         |                  | hard                                    | high            | hard               |

in flight vehicle trajectory optimization. Its method research is always about solving the “curse of dimensionality” of optimal control problem. Luns et al. [32] developed a dynamic programming of better performance: Iterative Dynamic Programming (IDP)). This method does not need to compute gradient information. Thus, IDP has a relatively effective performance in smooth and non-smooth optimal control problem. IDP solved the problem of expanding grid. IDP has been successfully applied in many practical optimal control problems, for example problems in chemical engineering field. And some high dimensional problems have been resolved. However, it has relatively bad performance in resolving problems like the online optimal control of air traffic and unmanned vehicle trajectory planning. Traditional dynamic planning and IDP have a common problem, that is the grid points in each stage of system are hard to be determined. As everyone knows, the more grid points are, the higher the accuracy is. At the same time, the time needed and storage space will increase with grid points by index number. Ref. [33] comes up with the dynamic planning method of single grid applied in trajectory optimization. This method can be used to solve some online optimal control problem, such as collision and avoidance of flight vehicles, the control of unmanned flight vehicles and optimal trajectory transference.

### 3.4 Rapidly-exploring random tree (RRT) method

Recently, some methods used in robot trajectory planning have been introduced to the solution of online planning and guidance of flight vehicle trajectory. RRT is a rapidly trajectory planning method which has been specifically introduced to online trajectory and demand optimization. The basic principle of this method is that creating two status trees, one starts from initial state and does forward integral; another starts from objective state (terminal) and does backward integral. Each step of status trees' growth is to do the forward integral or backward integral to flight vehicle dynamics. Accordingly, control variables are calculated from RRT calculation which can connect the two trees more quickly. When the two trees approach each other in a certain space distance, they can be connected as an available trajectory [34].

Assuming  $x_{init}$  is the initial point of planning and  $x_{goal}$  is the terminal point. RRT calculation can be described as follows [35]:

(1) Initializing propagating tree  $T$ . Taking the given initial point  $x_{init}$  as the root node  $t_1$  of tree  $T$ .

(2) Taking  $x_{dir}$  as the exploration direction point of the tree, using probability  $p$  to choose target point  $x_{goal}$  as the exploration direction point of the present tree, using probability  $1-p$  to choose a random point  $x_{rand}$  in the search space as the exploration direction point of the present tree.

(3) Choosing the tree growing point  $t_{grow}$ . Assume the present tree  $T$  has nodes of  $m$ , which is  $T =$

$\{t_i | i = 1, 2, \dots, m\}$ .  $d_i = \|t_i - x_{dir}\|$  is the distance between node  $t_i$  and  $x_{dir}$ . Choosing node  $t_{near} = \arg \min_{t_i \in T} \{\|x_{dir} - t_i\|\}$  as

the tree growing point  $t_{grow}$ , which is mostly close to exploration direction point  $x_{dir}$ .

(4) Calculating exploration node  $t_{new}$ . Assume  $S$  as the step length of exploration, then the new node of exploration  $t_{new}$  can be worked out by the following formula:

$$t_{new} = \begin{cases} x_{dir} & \dots \dots \dots \|x_{dir} - t_{grow}\| \leq s, \\ t_{grow} + s \cdot \frac{x_{dir} - t_{grow}}{\|x_{dir} - t_{grow}\|} & \dots \dots \dots \text{otherwise.} \end{cases} \quad (25)$$

(5) Updating propagating tree  $T$ . If  $t_{new}$  does not have clash with any obstacles, then  $t_{new}$  is added into the propagating tree, the propagating tree renews as  $T' = T \cup t_{new}$ ; or, gives up this exploration, that is to say  $T' = T$ .

(6) Estimating if the objective is achieved or not. If the new exploration point  $t_{new}$  satisfies  $\|x_{grow} - t_{dir}\| < \theta$ , among which,  $\theta$  is the preset threshold value, then supposing objective  $x_{goal}$  is achieved; if not, backing to step (2).

(7) Returning from target point to tree root node, and backing to path.

It is the RRT calculation that puts many constraints on control variable, for example, rate of change of maximum inclining angle and acceleration limits can be worked out in corresponding controlled space, simultaneously the branches of status tree, which are out of state variable constraints will be automatically removed. Therefore, the space control of state variable and control variable can be achieved automatically. It is one of the most significant advantages of the RRT method. It has a characteristic which we do not expect, that is, randomness. As to the on-line application, the randomness may make it impossible to get an available trajectory in a given time interval. Especially when the scale of problems gets larger, the degree of difficulty will rise sharply. The bottleneck of this method is how to determine the convergence metric for the complex dynamic systems. Such as, there are two state variables, flight-path angles and speed in the reentry problem. It is almost impossible for us to decide which state is closer to target unless we know how to get the flight vehicle to the target point under each state variable. This problem equals the design of overall state trajectory.

Ref. [36] realized the trajectory generation of auto-aircraft. Ref. [37] tried to use RRT method to solve the reentry problem. This method shows a good application prospect when solving some difficult trajectory planning problems such as helicopter trajectory planning [37]. The random-tree method can work out more than 10 available trajectories which can avoid several obstacles in a few seconds while the other optimization methods cannot obtain available trajectories under so many state constraints.

### 3.5 Rolling horizon procedure optimization method

Rolling horizon procedure optimization method draws on the idea of predictive control to optimize the flight vehicle trajectory on-line by finite segment method. Predictive control is a calculation of optimization control specifically invented to solve the inadaptability of the traditional optimization control to the industry process. Its main idea is to use the moving finite segment optimization to replace the regular overall optimization.

When introducing the idea of predictive control to optimization of flight vehicle trajectory, it is necessary to take the maximum optimization of terminal merit function into consideration. Therefore, it is necessary to solve the description of cost-to-go function while using moving time horizon optimization method. Usually, an approximate model is used to describe the function.

Many methods have turned out to solve this problem in predictive control [38]. Jadbabaie and others represented that the terminal merit function can be calculated by control Lyapunov function (CLF). As to the optimization trajectory of moving time horizon, it is regular to use the approximate trajectory model to calculate the surplus value function. For example, ref. [39] used polynomial to indicate two-dimensional terrain model and took minimum distance from approximate horizon target as the terminal merit function of RH planner. The approximate model can be calculated off-line according to the information on terrain and target. The mathematical description of rolling horizon procedure mixed integer linear programming is [40]:

$$\begin{cases} \min_{x_k, u_k} J_T = \sum_{j=0}^{T-1} (q' |x_{k+j|k} - x_f| + r' |u_{k+j|k}|) + s' |x_{k+T|k} - x_f|, \\ \text{s.t. } x_{k+j+1|k} = Ax_{k+j|k} + Bu_{k+j|k}, \forall j = 0, \dots, T-1, \\ x_{k+0|k} = \hat{x}_{k|k-1}, \\ u_{k+j|k} \in U, \forall j = 0, \dots, T-1, \\ p_{k+j|k} \notin U, \forall j = 0, \dots, T, \end{cases} \quad (26)$$

where  $T$  indicates the length of controlling and predicting (planning) time-domain.  $x_{k+j|k}$  indicates the prediction of state of system at the moment of  $k+j$  under the comprehensive effect of dynamic equation of system, state  $x_k$  at moment  $k$ , and the maximum optimization control sequence  $\{u_{k+j|k}\}_{j=0, \dots, N-1}$  when it is at discrete moment  $k$ . Performance index function  $J_T$  requires that the state of flight vehicle should turn into terminal state  $x_f$  as soon as possible, and simultaneously takes the control of  $u$  and the performance requirement of the terminal state  $x_f$  into consideration, among which,  $q$ ,  $r$  and  $s$  are weight vectors of corresponding items.

There are some uncertainties in the actual application of trajectory optimization. For example, the target might have changed before the flight vehicle reaches the destination,

and the details of the environmental parameter are also hard to know well and truly beforehand. When the trajectory is longer, there will be more uncertainties. Rolling horizon procedure optimization optimizes trajectory in infinite segment, so it has superiority for dealing with uncertainties and it is in favor of realizing the real-time planning of trajectory.

At present, this method is mainly applied to the trajectory planning of low altitude flight vehicle. Such as the two-dimensional or three-dimensional real-time planning of terrain following flight vehicle trajectory; the on-line replanning of the flight vehicle trajectory under the circumstances of the changed threatening, terrains, and targets [41–43]. This method combines with the off-line optimizations, such as dynamic planning, to realize the real-time trajectory optimization.

### 3.6 Genetic algorithm

Genetic algorithm (GA) is an intelligent overall optimization calculation method independent of gradient information which simulates natural selection and genetic mechanism. It is robust, not sensitive to initial value, and has good performance dealing with the overall optimization of some gigantic and complex nonlinear system. GA is a widely used search method based on the theories of natural selection and population genetics. GA search method combines the Darwin's thought of the survival of the fittest and random information exchange. The thought of the survival of the fittest removes the known inadaptabilities and random information exchange uses the known knowledge, therefore the search process has been improved.

Basic thought of GA. GA begins with a population which represents the problem that may have a potential solution set. A population is made by individuals of a certain number which has been coded genetically. According to the theory of survival of the fittest, the original population will be evolved into better approximate solution after its birth. Individuals are selected by individual adaption in problem domain at each generation, with the help of crossover and mutation based on genetic operator from natural genetics to reproduce a population which represents a new solution set. This progress will cause the naturally evolved offspring population to become more adaptive to environment than its ancestors. The elitists among the last population can be regarded as the approximate optimal solution after being decoded [44].

Simple GA uses fixed length of base  $n$  symbol string to represent the individuals among a population, the alleles of which are composed of symbol set  $\{0, 1, \dots, n\}$ . The genic value of each individual among the original population can be generated by uniformly distributed random numbers [45].

The evolution of organisms is achieved by the cooperation of the crossover of each chromosomes and the disorder of chromosome. Accordingly, the search process of optimal

solution in GA is also to simulate the evolution of organisms, using the so-called genetic operators to act on population  $P(t)$  to get a new population  $P(t+1)$  by the following genetic manipulation.

The main genetic operators are as follows:

(1) Operators' selection. Selecting some superior individuals from population  $P(t)$  of generation  $t$ , genetically transmitting themselves to the next population by certain rules and methods according to individual fitness. The individual selecting probability varies according to the individual fitness. The individuals that are weak on fitness have small probability to be chosen while the others which are strong on fitness have large probability. It reflects the thought of survival of the fitness.

(2) Crossover operators. Pairing individuals among a population randomly and exchanging part of chromosome inside each pair according to a certain probability (named probability of crossover  $P_c$ , crossover rate). Taking the single-point crossover of a binary coded genetic algorithm for example, the method of which is as follows:

$$\begin{array}{lcl} A = (0110, 1100) & \text{Crossover} & A' = (1110, 1100), \\ B = (1110, 0100) & & B' = (0110, 0100). \end{array}$$

(3) Mutation operators. Changing the gene value for other alleles on one or more gene locations of each individual in population  $P(t)$  by a given probability (named mutation rate  $P_m$ ). Taking binary genetic algorithm code for example:

$$A = (01\bar{1}10001) \text{ Mutation} \Rightarrow A = (010\bar{1}0001).$$

The basic algorithm process is described in Figure 1. From the figure, we could know that GA demands group operation. Besides the three mentioned genetic operators, designs on population size, fitness function and control parameters are necessary in operations.

Because of the different kinds and the enlarging scale of problems, a universal method which can solve the problem of search and optimization at limited cost is searching. GA is an effective method which is different from traditional methods of searching and optimizing. The main distinctions are: (1) self-organization, self-adaptive and autonomous learning; (2) essential parallelism; (3) GA does not require

guidance or other knowledge assistances, but requires the objective function which influences the direction of searching and related fitness function; (4) GA focuses on probability transition rules, not on fixed transition rules; (5) more direct application than that of other methods; (6) GA can generate many potential solutions for given problems, which will be chosen by decision.

Effective using of global information and implicit parallelism are two main characteristics of GA which has strong global optimization because it lays less limits on problems. But GA tends to premature convergence which will cause failures in copying effective genes from other individuals and lead to gene loss. And at the later stage of evolution, the minimum difference among chromosomes, stagnation of the evolution of the whole population and low search efficiency will cause unsuccessful searching for global optimal solution.

Since Holland systemically proposed integrated configuration and theory of GA from 1975, many scholars have committed themselves to promote development of GA, and thoroughly researched into coder mode, confirmability of control parameters, selection fashion and crossing mechanism and so on. Basic approaches which are generalized are as follows: (1) changing the component part or application techniques of GA; (2) adopting hybrid GA; (3) adopting dynamic adaptive techniques, adjusting the key parameter of algorithm and particle size of code during the evolution; (4) adopting nonstandard genetic operators; (5) adopting parallel genetic operator and others.

The earlier reports on the application of GA to optimization design of flight vehicle trajectory came out around 1996, American scholars adopted basic GA to study the problem of low-thrust trajectory of flight vehicle flying outside atmosphere [46, 47]. Henceforth more and more documents of applications of GA to flight vehicle trajectory transfer and planetary exploration trajectory optimization have appeared [48, 49], and focused on the application of the improved GA to trajectory optimization. In about 1999, GA was applied to optimize reentry trajectory for launched vehicle after the success in its application to flight vehicle trajectory transfer [50, 51]. Since then, GA has become a research focus and its latest achievements have been widely used.

The development in China resembles development in other countries, but a bit slower. The earlier work is about the minimum-thrust orbit transfer of lunar satellite [52], and gradually it had been applied to the return trajectory of the reentry flight vehicle [53, 54] and the trajectory optimization design of launched vehicle [55]. Ref. [56] proposes to solve the problem of optimal glide trajectory by GA based on adaptive mutative scale chaos. The results verified that the improved GA could be applied to solve the problem of trajectory optimization. Ref. [57] uses GA to optimize and design the deep space exploration trajectory. Ref. [58] proposes a method that uses chaos factors, concentration of

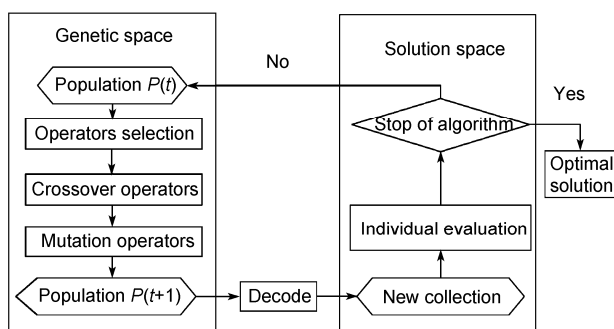


Figure 1 Description of the basic algorithm process.

immune mechanism and adaptive vaccine generation. This method can improve GA. It uses the randomness, ergodicity and internal law of chaos to initialize population and reduce the blindness in searching, so the initial population can be evenly distributed in the solution space. Meanwhile, it uses concentration of immune regulating mechanism to maintain the population diversity and avoid premature phenomena. Then it introduces the adaptive vaccine generation to speed up optimizing, raise the rate of convergence and improve the optimization; and the algorithm is applied to trajectory optimization of aerospace vehicle, regarding the minimum fuel consumption as the optimization goal during the climbing course. The availability of this method has been proved by simulation.

### 3.7 Ant colony algorithm

In 1991, the Italian scholar Colomi et al. [59] presented a new swarm intelligent bionic optimization algorithm—ant colony algorithm (ACA), which was based on the collective behaviors of foraging, task distribution and cemetery construction of ants. It has been used to solve traveling salesman problem successfully. It is a universal bionic heuristic algorithm based on the behavior of populations tracking the optimal solution. Depending on the foraging behavior and strategy of ants it uses the motion path of ants to represent the initial solution of solving problem. Independent of the mathematic expression of solving problem, it has been optimized to find the global optimal solution. This algorithm has many advantages, such as positive feedback, well robust performance, excellent distributing parallel computation system, and easy to work with other methods. But it also has some disadvantages, such as overtime searching, tendency towards local optimization, and slow convergence [60].

ACA has two basic phases [61, 62]: adaptive phase and cooperative phase. At adaptive phase, each candidate solution adjusts self-structure according to the accumulated information; at cooperative phase, each candidate solution exchanges information expecting to generate more functional solutions, which resembles the learning mechanism of the learning automata.

To get a better understanding of the basic theory of ACA, the classic symmetric TSP problem is usually used to do explanation. The specific steps of the basic ACA are as follows [60].

(1) Parameter Initialization. Setting maximum cycle number  $N_{\text{cmax}}$  with time  $t=0$  and cycle number  $N_c=0$ , putting  $m$  ants on  $n$  cites, ( $m < n$ ). Making initialized information content  $\tau_{ij}(t) = \text{const}$  of each edge  $(i, j)$  on the directed graph,  $\text{const}$  indicates constant, and initial time is set ( $\Delta \tau_{ij}(0) = 0$ ).

(2) Cycle number  $N_c \leftarrow N_c + 1$ .

(3) The number of ants  $k \leftarrow k + 1$ .

(4) According to the probability worked out by state

transition probability formula, individual ant chooses city  $j$  and proceeds.

(5) Revising the pointer of tabu table, that is to say, moving the ants to new city after selection, and moving the city to the ant's tabu table.

(6) If the cities of the set have not been experienced, that is to say,  $k < m$ , then backing to step (3), if not, then going to step (7).

(7) Updating information content on each path.

(8) If the terminate requirements are satisfied, that is if cycle number  $N_c > N_{\text{cmax}}$ , then ending the cycle and outputting the result calculated by program. If not, clearing up tabu table and backing to step (2).

As a swarm intelligent bionic heuristic algorithm, ACA itself has advantages and disadvantages [63]. Its main advantages are:

(1) Distributed computation. ACA is a distributed multi-agent system which can independently and simultaneously search solution at different spots in the problem space. The process of solving problem would not be influenced by any ants' failures. This enhanced not only the reliability but also the global searching ability.

(2) Self-organization. Self-organization refers to the organization or the organizational instruction from the inside of system, which means the disorder-order transition of the system without the outside interference. ACA reflects this characteristic. At the initial state of algorithm, ants search for solution solely and disorderly. After a period of evolution, ants tend to search some approximate optimal solutions. This not only embodies the evolution from disorder to order of self-organization, but also enhances the robustness of algorithm, so the information about solving each aspect of problem is not necessary to be known while applying the algorithm. Therefore, it can be applied to problems of a kind.

(3) Positive feedback. Positive feedback refers to enhancing the future behaviors by present behaviors. The accumulation of ant's pheromones reflects the process of positive feedback: leaving more pheromones on the path passed by better solutions and pheromones draw more ants. This process enlarges the initial value and leads the whole system to evolve to the direction of optimal solution.

(4) Strong robustness. The solution of ACA does not rely on the initial path and it does not need manual adjustment while searching. It can apply to other problems with a little adjustment on its model. Without the center control and data, it would not be influenced by any ants' failures.

(5) Easy to work with other methods. It is easy for ACA to work with various heuristic algorithms, such as genetic algorithm, immune algorithm, particle swarm algorithm, to improve the performance of ACA.

Its main disadvantages are [43, 44]:

(1) Overtime searching. Comparing with other intelligent bionic optimization algorithms, it takes more time to search.

(2) Having tendency towards local optimal solution. All

the solutions found by each individual are approximate and it cannot make further search in the solution space when the search reaches a certain state. It does not avail finding a better solution, this is named stagnation behavior.

Ref. [64] adopts ACA to study the optimization of lunar soft landing trajectory. To enhance the searching accuracy, the local search strategy should be added into decimal ACA which is an improved ACA adaptive to continuous parameter optimization. Then it is applied to the optimal trajectory design of lunar soft landing. Ref. [65] uses the discretization to deal with the continuous searching space, introduces pheromone diffusion mechanism and proposed method of adoptive search space adjustment, and applies this improved ACA to the practical problem of trajectory optimization. Because of the superiority of ACA on solving complex optimization problems, especially discrete optimization problem, many scholars [66–68] have applied it to flight vehicle path planning today and it shows its superiority.

ACA is a new bionic evolutionary algorithm whose theory is relatively simple and it has already expressed its characteristics and charms. But its theory is far beyond maturity comparing to some other theories such as genetic algorithm theory and tabu search algorithm theory, its potential needs further exploration and there are many challenging problems to be solved. Main aspects are shown below [69].

(1) Putting more emphasis on the basic theory research and further development on new mathematical analysis and modeling tools are important. Especially the basic theory research is very important which includes the estimation of convergence and its rate when solving various optimizations, avoiding stuck in local optimal value, analysis on robustness and the design theory of parameters  $\alpha$ ,  $\beta$ ,  $\zeta$ ,  $\rho$ ,  $Q$ . At present, the discussions on the number of ants and the mathematical analysis on operating time of the algorithm remain unfound. Therefore, the study on the theory of mathematical basis of the algorithm will be an important subject in the future research.

(2) It is possible to combine ACA and other methods to develop hybrid optimization, and then to develop the more powerful intelligent behaviors which can deal with more complex systems with advanced ideas [70–73].

(3) The new theory must be examined by practice. New problems can be found during practice, the practice therefore improves the theory.

(4) Besides, the hardware realization of ACA will become one of the research focuses.

The profound researches on problems above will greatly contribute to development of the application and theory of ACA which will have a bright future in the field of intelligent computation.

### 3.8 Particle swarm optimization

Enlightened by the findings of artificial life study, particle

swarm optimization (PSO) is an evolutionary computation which was first represented in 1995 by psychologist Doctor Kennedy and Doctor Elberhart working on computational intelligence based on Swarm Intelligence [74, 75]. It quickly drew attentions from academia and soon was regarded as a prospective optimization algorithm because of its simple concept, easy realization and rapid convergence. Presently it has been applied to many fields such as multi-objective optimization, pattern recognition, signal processing and decision support; and was listed as a topic in international evolutionary computation conference.

The operation principle of PSO is independent of the natural law of individual evolution. Based on the study of bird's foraging behavior, it simulates the social behaviors of biotic population. The interactions between individual and individual, individual and population exist in biotic population, which represents an information share mechanism in biotic population. PSO is a simulation of this social behavior, which uses the information share mechanism to improve the progress of the whole population by exchanging experience between individuals [76].

PSO reproduces the initial population. It randomly initializes a particle swarm in the space of feasible solution; each particle is a solution to optimization and it gets a fitness value by objective function. Each particle motion in the space of solution and its direction and distance are determined by a certain speed. Usually particles would follow the motion of the present optimal particle and search each generation to get the optimal solution. Particles track two optimal solutions in each generation, one is the particle that has ever been found, another is the whole population that has ever been found [76].

Particle calculates the new speed of itself by tracking the global extremum and the local extremum. Particles move to a new place. The superiority and inferiority of each particle are judged by the defined fitness function, which is related to the solved problem [76].

The main process of PSO is as follows:

(1) Initializing particle swarm, including population size, the position and speed of each particle; setting  $c_1$  and  $c_2$  as acceleration coefficients, and maximum evolution algebra  $T_{\max}$ . Setting the present evolution algebra  $t=1$ , randomly generating  $m$  particles  $X_1, X_2, \dots, X_m$  in definition space  $R$  to constitute initial population  $X(t)$ ; the random displacement  $V_1, V_2, \dots, V_n$  of each particle constitutes displacement matrix  $V(t)$ .

(2) Calculating each particle's fitness value of each dimension.

(3) Comparing the fitness value with pbest of every single particle. If the present value is better than pbest, then setting pbest as the present value and setting the position of pbest as the present position in  $n$  dimension space.

(4) Comparing fitness value of each particle with population optimal value. If the present value is better than gbest, then setting gbest as matrix's subscript and fitness value of

the present particle.

(5) Renewing the speed and positions of particles, and generating a new population.

(6) If the terminal requirement is achieved (the deviations are enough or reaching the maximum cycle number), then exit, if not, back to step (2).

The main advantages of PSO [77] are: (1) easy to describe, (2) less parameter, (3) easy to realize, (4) quick convergence.

PSO can be achieved easily with a low computation cost and less occupancy of computer hardware resource. PSO has been proved to be a good solution to many global optimization problems. Of course, PSO, like other global optimization algorithms, is subject to the defects of local optimal, low convergence precision and slow convergence velocity in later period.

Ref. [78] tests respectively the initial mass, starting angle and horizontal displacement by adopting the improved PSO and performs a global optimization on the flying trajectory referring to the test results; ref. [79], taking reentry optimization of hypersonic flight vehicle with minimum control energy as an example, simplifies flight vehicle motion model, parameterizes the controlled quantity, and deals with the constraint by transformation processing. PSO introduces Adaptive Weight and sets the PSO initial parameters and weighting by full utilization of flight vehicle's motion characteristic in reentry, which makes the calculation time short, and at the same time, guarantees the convergence of the algorithm. It also analyzes and compares the emulation-al steps' influence on the results. Ref. [76] uses PSO with multiple parameters to study the relationship between flight vehicle's main parameters: initial quality, starting angle, wing area, lift drag ratio and horizontal displacement. Besides, it performs optimization on flying trajectory of single period of skip flying by referring to the research results. Furthermore, it also performs optimization on skip flying trajectory with uniform parameters design on the basis of PSO. Ref. [80] studies the ascent flight trajectory optimization design problem of solid carrier rocket by applying PSO.

PSO is a relatively new evolutionary algorithm based on swarm intelligence, the study of which still lacks of solid theoretical basis and systematic analytical methods. Many problems still call for further study [77].

(1) The theories foundation research of PSO is still quite sparse, lacking deep-going theoretical analysis with universal meaning. However, any algorithm desiring for a long-term development must need a solid theoretical basis. Therefore, in-depth theoretical analysis on the algorithm is still needed.

(2) Selection and design of parameters of algorithm. Parameters of algorithm are the key factor which influences the performance and efficiency of the algorithm. Different parameters are needed in different problems. However, now there still has no universal approach to select the optimal parameter. Thus those parameters of many practical prob-

lems only can be selected from users' experience. The classification of disparate problems and the universal approach to select parameters is also one of our research directions.

(3) Many researchers classify PSO as one of the evolutionary algorithms. Like other evolutionary algorithms, PSO also will fall into local optimal. Therefore, to improve PSO from perspectives of introducing multiple populations, modifying topological structures of populations and combining with other optimization algorithms are also one of our research directions.

(4) Because of PSO's inherent parallelism, parallel computing of PSO will be an important research direction.

(5) The application area of PSO has not yet been broadened. PSO has been applied successfully in evolutionary neural network. However, its application in many other fields still stays at research stage. The application fields of PSO need to be developed and summarized.

### 3.9 Dynamic programming and conjugate gradient combinational algorithm

#### 3.9.1 Dynamic programming

Flight vehicle trajectory is a four-dimensional flight vehicle trajectory. It is represented by four parameters: location ( $x, y, z$ ) and flight time  $t$  or velocity vector  $V(V_x, V_y, V_z)$ . Flight vehicle trajectory will be optimized in a grid coordinate system ( $o-\varphi r\theta$ ).  $o$  is geocenter,  $\varphi$ , longitude,  $r$ , the distance between flight vehicle and geocenter,  $\theta$ , latitude. All of the points on flight vehicle trajectory will be presented by  $P(i, j, k)$ .  $i, j, k$  represent respectively the values of  $\varphi, r, \theta$  directions in the grid coordinate system. The performance index at point  $P(i, j, k)$  is defined by the following formula [78–80]:

$$J_{\{i,j,k\}} = \min_{T(i,j,k)} (x_{(i,j,k)}, T_{(i,j,k)}, J_{(i,i+1)}), \quad (27)$$

where  $J_{\{i,i+1\}}$  is the performance index between  $i$ th and  $(i+1)$ th levels.

$$J_{\{i,i+1\}} = \min_{T(i,j,k)} (x_{(i,j,k)}, T_{(i,j,k)}, J_{\{i_1,j_1,k_1\}}), \quad (28)$$

where  $i_1 = i + 1, j_1 = j + 1, k_1 = k + 1$ , performance indexes of all the terminals are to be

$$J_{(N,j,k)} = 0, (j = 1 - N, K = 1 - N), \quad (29)$$

where  $N$  is the maximum of  $i$  (the final position), therefore, the general performance index  $J_{\{1,j,k\} \leftarrow (N,j,k)}$  is minimum:

$$J_{\{1,j,k\} \leftarrow (N,j,k)} = \min_{j,k} [J_{\{1,j,k\}}]. \quad (30)$$

The specific steps are as follows:

**Step 1.** In Descartes Cartesian coordinates ( $o-xyz$ ), the grid coordinate system ( $o-\varphi r\theta$ ) is set according to start point and target point of the flight vehicle. The size of the

grid  $(\Delta\varphi, \Delta r, \Delta\theta)$  is determined by mobility of the flight vehicle for the engine thrust is restricted ( $T \leq T_{\max}$ ).

**Step 2.** Calculate flight vehicle mobile space (passage)

$$\begin{cases} \varphi_{\min} \leq \varphi \leq \varphi_{\max}, \\ r_{\min}(\varphi') \leq r(\varphi') \leq r_{\max}(\varphi'), \\ \theta_{\min}(\varphi') \leq \theta(\varphi') \leq \theta_{\max}(\varphi'). \end{cases} \quad (31)$$

The flight space corridor becomes larger and larger because the flight vehicle quality is reduced and the working time of the engine is increased, which makes the flight vehicle mobility to be strengthened.

**Step 3.** Minimize the performance index

$$J_{\{1,j,k\} \leftarrow (N,j,k)}.$$

**Step 4.** According to Step 3, from  $i = i_{\max} = N$  to  $i = i_{\min} = 1$ , storing all the performance indexes of every point  $J_{\{i,j,k\}}$  and relevant parameters (flight trajectory, control law, and flight vehicle mass).

**Step 5.** On the  $i_{\min}=(1)$ th level, finding the flight trajectory of the maximum performance index from matrix  $J_{\{1,j,k\}}$ . It will output all of the results, such as the optimal flight trajectory  $(t, x, y, z, V_x, V_y, V_z)|_i$ , ( $i = 1, 2, \dots, N$ ), optimal control law, flight vehicle mass and other parameters.

### 3.9.2 Conjugate gradient method

On account of the discrete flight trajectory  $(t, x, y, z, V_x, V_y, V_z)|_i$ , ( $i = 1, 2, \dots, N$ ), two-point boundary value algorithm in optimal control theory is used to optimize trajectory and get the accurate flight trajectory  $(t, x, y, z, V_x, V_y, V_z)$ . The calculation of accurate optimal flight trajectory section  $(t, x, y, z, V_x, V_y, V_z)^*|_i$  in the  $i$ th level is achieved by the two consecutive points  $(t, x, y, z, V_x, V_y, V_z)|_i$  and  $(t, x, y, z, V_x, V_y, V_z)|_{i+1}$ . The calculation of accurate optimal flight trajectory section  $(t, x, y, z, V_x, V_y, V_z)^*|_i$  in the  $i$ th level can be obtained by solving the following trajectory optimization problem [81–83].

Minimizing the performance index of the accurate optimal flight trajectory in the  $i$ th level:

$$J_i = \min_{u_i(\eta, \varepsilon_T, \sigma)} m_f, \quad (32)$$

where the subscript  $i$  represents the numerical value of the flight trajectory in the  $i$ th level.

The locations of start point and terminal point are:

$$\begin{aligned} \mathbf{x}_i(t_0) &= (t, x, y, z, V_x, V_y, V_z)|_i, \\ \mathbf{x}_i(t_f) &= (t, x, y, z, V_x, V_y, V_z)|_{i+1}. \end{aligned} \quad (33)$$

Since there is still no global trajectory optimization algorithm in true sense, this combinational algorithm, cultivating the strengths and avoiding weakness, takes dynamic programming as the suboptimal solution meeting certain requirements in the first phrase. Then it solves the two-point boundary value problems by applying the conjugate gradient method. Through  $N$  times calculations, the optimal flight trajectory in section  $N$  can be obtained. The whole optimal trajectory includes 4-D optimal flight trajectory  $(t, x, y, z, V_x, V_y, V_z)^*$ , the control variate  $U^*$  and the remaining mass of flight vehicle  $m_f$ . This combinational optimization algorithm is a global optimization for the optimal flight vehicle trajectory.

Ref. [81], regarding the minimum fuel consumption as the performance index, taking various constraint conditions into account (overload, dynamic pressure and heating rate), optimizes the ascent trajectory of the aerospace planes by applying this combinational optimization method. Ref. [82], aiming at optimization design of low thrust four-dimensional track in deep space exploration, performs a global optimization on four-dimensional track for deep space exploration on the basis of two-body model by applying this combinational method. Ref. [83] performs a global optimization on low thrust and multiple target four-dimensional track for deep space exploration.

## 4 Hypersonic vehicle trajectory optimization review

With the development and consummation of hypersonic aeronautical technology, the research currently focuses on hypersonic vehicle to achieve a long-distance cruise and global reach, i.e. reaching any point on the whole surface of the Earth. Hypersonic vehicle refers to the vehicle whose flight Mach number is greater than 5. There are some unique advantages for this hypersonic vehicle, such as high flight speed, strong penetration capability, global reach and great power to damage, which makes it become today's hot and focal spot of weapons development in countries all over the world [84].

Hypersonic cruising vehicle shows its significant meaning of development and military value at least in four aspects [85]:

(1) Hypersonic cruising vehicle can overcome the deficiencies of subsonic scheme in penetration capability and destructive capability to destroy mobile missile launchers and reinforced targets in deep underground.

(2) Compared with other weapons executing the same kind of tasks, hypersonic cruising vehicle has smaller volume, more light weight, lower cost and is adequate for multiple platforms launching with various attacking ways and broad applications. Therefore, it has greater military and economic efficiency.

(3) In the sky above 24 km height, if taking a hypersonic maneuver flight with Mach numbers 4–8, it will be difficult for the existing surface anti-vehicle weapons and airborne intercept weapons to effectively intercept hypersonic cruising vehicle and even the American Missile Defense System will lose its due defense power, because the interception efficiency distinctly drops.

(4) With the scramjet engine technology getting to be more mature day by day and with the successful development of new hydrocarbon fuel, not only scramjet engines can be ignited and started to combust at a flight speed with Mach numbers 3.5–4.0, but also it breaks through the traditional limitation to supersonic and hypersonic flight which makes the start point of hypersonic flight shift down to Mach number  $\gg 4$  from Mach number  $\gg 5$ . This alteration will not only affect the classification principle and standards of missile with air-breathing engine as power system, but also will greatly change the supersonic and hypersonic missiles in design concept and structural features.

In a high technological and informationalized war in the future, hypersonic cruising vehicle will play a more and more important role and become the main part of attack and fight system.

On account of the long flight time and wide flying range of hypersonic vehicle, its trajectory in a vertical plane is divided into ascent section, lead-in section, sliding section and descent section in accordance with different features of each section [86].

The optimization of ascent section trajectory is an essential and basic issue regarding to hypersonic vehicle study, which can reduce launched weight or lower the cost to greatest extent, or reduce the requirements to each subsystem (propulsion system, materials and pneumatic appearance) as well as provide each subsystem with necessary data, which has a great theory and engineering significance. The mathematical model of the flight trajectory is represented as follows [87]:

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, I_{sp}, m, t) \quad (34)$$

where  $\mathbf{x} = [V, \theta, \psi, x, h, z]$ ,  $\mathbf{u} = [\alpha, \sigma, P]$ ,  $V$  is the speed of the vehicle,  $\theta$  is the path angle of the trajectory,  $\psi$  is the yaw angle,  $x$ ,  $h$  and  $z$  are flight horizon, height and lateral direction distance respectively;  $\alpha$  and  $\sigma$  are angle of attack and roll angle of the control variables,  $P$  is the thrust of unit weight,  $I_{sp}$  is the specific impulse of the engine,  $m$  is the vehicle mass.

The border terms and performance index are as follows:

1) Initial boundary

$$\mathbf{x}(t_0) = \mathbf{x}_0. \quad (35)$$

2) Terminal boundary

$$h(t_f) = h_f, \quad \theta(t_f) = \theta_f. \quad (36)$$

3) The constraints of attack angle and its change rate

$$|\alpha(t)| \leq \alpha_{\max}, \quad |\dot{\alpha}(t)| \leq \dot{\alpha}_{\max}. \quad (37)$$

4) The optimal performance index is to maximize terminal gross energy

$$J = - \left[ \frac{V^2}{2g} + h \right]. \quad (38)$$

In the 1990s, Calise [88] and Schmidt [89] studied the ascent trajectory of hypersonic vehicle. They took the vehicle modeling as the mass point and divided flight dynamic into fast time variation and slow time variation by applying singular perturbation to obtain the suboptimal trajectory which saves most fuel. The limitation of this method lies in that it cannot effectively solve the state-constraint. Lu et al. [90] and Dukeman [91] also studied the guidance in ascent section in recent years. Aiming at vehicle driven by rocket, Lu discussed the optimal ascent guidance problem of RLV in endoatmosphere. However, Dukeman tried to solve the two-point boundary value problem by approximate simplification forms equations and the application of multiple-point shooting method. Li et al. [92] proposed a feasible scheme to solve the optimal ascent trajectory on account of features of hypersonic vehicle. On the assumption that yaw angle is zero degree and the moment of force gets transient equilibrium, performing optional modeling on the flight issue in ascent section, introducing the mass as quality of state and taking saved most fuel as the index and taking direction of the thrust as the option controlled variable, the optimal condition in first phase is deduced according to the maximum principle. The optimal trajectory can be solved by solving the two-point boundary value for the numerical value, taking analytical solutions as initial conjecture, applying the classical finite difference method and improved Newton method and homotopy method under the condition of meeting the angle of attack process constraints. Stimulation is conducted under the restriction of initial constraint and terminal constraints. Ma et al. [93] built the zero dimension mathematic model of optimization on hypersonic vehicle trajectory in endoatmosphere and proposed an improved adjoint method to solve the optimal two-point boundary value. The astringency of their algorithm also has been proved. When it was applied to the optimization of ascent section trajectory of hypersonic vehicle driven by solid-propellant rocket engine in dense atmosphere, under the premise of bending moment constraint, a group of slide controlled variables could be obtained indirectly.

The lead-in section is a track of trajectory from the terminal of the last level boosting phrase to the beginning of sliding track. In this section, the trajectory is required to be led into the initial height, speed and path angle of sliding track under the condition of unpowered flight and minimum loss of speed as well as heat flow and overload constraint should be guaranteed. Mathematical description of the lead-in section is: The initial height  $h_{t_0}$ , Mach number  $Ma_{t_0}$ ,

path angle  $\theta_0$ , terminal height  $h_t$ , path angle  $\theta_t$  are given; the process of looking for a control (attack angle  $\alpha$ ) makes the terminal speed maximum. At the same time the aerodynamic heating rate  $\dot{Q} < \dot{Q}_{\max}$  and the overload  $n < n_{\max}$  should be guaranteed.

Zhou et al. [94] have studied the optimization of lead-in section of trajectory of hypersonic gliding vehicle. They transformed this flight trajectory problem into optimal control problem and obtained the essential condition of optimal flight trajectory in first phrase by the maximum principle. They also solved the two-point boundary value by genetic algorithm. They first found the changing rule of angle of attack from the suboptimal flight trajectory, then deduced the range of variation of the initial adjoint variable from the range of angle of attack estimated from suboptimal flight trajectory, and finally optimized the initial adjoint variable within this range by genetic algorithm to obtain the global optimized flight trajectory and corresponding initial adjoint variable.

The sliding section is the following section of lead-in section. We define gliding as a kind of flight of a vehicle which happens when the engine is shut off and the vehicle starts to fly with certain Mach number and a zero path angle. It has requirements for height and speed at terminal moment. And at the same time, the heat flow and overload constraint should be guaranteed. Mathematical description of the sliding section is given below.

The initial parameters (flight height  $h_{g0}$ , Mach number  $Ma_{g0}$ , path angle  $\theta_{g0}$ ) and terminal parameters (flight height  $h_{gt}$ , path angle  $\theta_{gt}$ , Mach number  $Ma_{gt}$ ) are set; the process of searching for a control (attack angle  $\alpha$ ) makes the gliding range maximum, and at the same time makes the aerodynamic heating rate  $\dot{Q} < \dot{Q}_{\max}$  and the over load  $n < n_{\max}$ .

The reference flight trajectory needed for gliding trajectory generally can be created by either on-line or off-line methods. By using off-line flight trajectory programming approach, a reasonable optimal reference flight trajectory can be projected, which is optimized by a great deal of trajectory optimized design calculation according to the given path constraint and terminal constraint before being launched. However, when the on-line method is used, due to the uncertainty in the flying environment and characteristics of nonlinearity kinetics, it will be hard to accurately track the reference flight trajectory merely depending on the controlled aerodynamic force of rolling angle, which results in a large error for longitudinal travel, horizontal travel and heading angle. Therefore, it is difficult to handle the uncertainties when this method is used with a poor flexibility and low tracking accuracy. The on-line reference trajectory programming approach can, according to the current flight conditions of the gliding vehicle, shorten the time of trajectory generated by a certain flight trajectory simplification planning strategy, work out a reference trajectory to meet all

of the constraints, then track and control it. Compared with off-line flight trajectory programming approach, the on-line method is more flexible and timely with stronger adaptivity.

Generally, the standard flight trajectory guidance method can separately guide lengthways and lateral guidance, and the lengthways guidance is dominating. The lengthways guidance follows the height-speed profile (H-V) and makes real-time lift force adjustment in accordance with flight conditions and requirements of operation tasks to meet the requirements of flight tasks. The lateral guidance guarantees the accuracy of heading angle by adjusting symbol of rolling angle on the basis of heading angle error corridor or horizontal travel deviation. In general, lateral control of gliding vehicle of space shuttle can be achieved under condition of a long horizontal travel by adopting heading angle error corridor method, in which the error of terminal heading angle is large and the guidance accuracy is low. The launched vehicles (RLV) can be repeatedly used and deviation distance method can be adopted to achieve lateral control under the condition of a short horizontal travel, in which the error of terminal heading angle is small. Before adopting the former method to conduct lateral control, the error corridor of heading angle must be confirmed by massive numerical experimentations or engineering tests, which needs a great amount of calculation and cannot be transplanted easily. While the latter method is only fit for short horizontal travel control and cannot handle the long horizontal travel control confronted by gliding warhead. Therefore, Wang [95], integrating the long horizontal travel features of gliding warhead, proposed an improved strategy of self-adapting lateral control. According to the high degree of nonlinearity in gliding section of hypersonic vehicle, ref. [84] proposes gradual segmentation parametric method optimization strategy, and obtains the suboptimal solution of the problem by SQP algorithm. And on that basis, it constructs an equivalent system of the original system and demonstrates the relationship between the costate variables of the two systems, and puts forward the segmentation shooting method along with the suboptimal flight trajectory. After the two-point boundary value problem (TPBVP) being solved, the optimal solution is obtained by transformation.

The descent section refers to starting from the end of gliding section to reach a certain path angle by control in certain space span, and making the flight time shortest and at the same time satisfying the constraints of heating flow and overload. Mathematical description of the descent section is: Minimize the flight time by optimal control (attack angle  $\alpha$ ) based on the initial condition (flight height  $h_{d0}$ , Mach number  $Ma_{d0}$ , path angle  $\theta_{d0}$ ) and terminal condition (flight height  $h_{dt}$ , path angle  $\theta_{dt}$ ). At the same time, the aerodynamic heating rate  $\dot{Q} < \dot{Q}_{\max}$  and the over overload  $n < n_{\max}$ .

Yong et al. [96] studied the quick algorithm of trajectory

optimization problem for hypersonic vehicle to glidingly reenter based on a new solution to optimal control problem: Gauss Pseudo spectral method (GPM). Aiming at overcoming the difficulties of trajectory optimization problem of gliding reentry under condition of long-distance multiple constraints, they proposed a serial segmentation optimization strategy including three aspects: (1) constructing the starter builder of design variable and obtaining approximate optimal solution as optimized starter; (2) proposing a serial segmentation optimization strategy from feasible solution to optimal solution; (3) introducing equilibrium glide conditions and constructing dynamic segment points, then dividing the reentry trajectory into initial descent section and gliding section to obtain the solution. Liu et al. [97] conducted an optimization design on attack angle in reentry braking section of hypersonic vehicle by application of genetic algorithm and penalty function strategy to maximize the terminal speed under condition of meeting each course constraint and paralleled flight constraints. Dong et al. [98] proposed a quick trajectory programming method to meet the real-time requirement of on-line trajectory programming for hypersonic cruising vehicle in glissade section. Based on mathematical model, hierarchical planning is adopted. Firstly, transforming the flight trajectory and quasi-equilibrium constraints into safe corridor, then working out the standard flight trajectory in safe corridor according to terminal constraints and centre line principle and applying PRP (Polak-Ribiere-Polyak), the conjugate gradient method, to optimize the standard flight trajectory to obtain optimal flight trajectory based on the corresponding control sequence of standard flight trajectory, simulating the widest flight range and flight trajectory in shortest time. Xie et al. [99] proposed to apply PSO to reentry optimization of vehicle. They took optimization of the reentry of hypersonic vehicle with minimum control energy as an example to simplify vehicle motion model, parameterize the controlled quantity. PSO adopts adaptive weight and sets the PSO initial parameters and weighting through full utilization of vehicle motion characteristic in reentry, analyzing and comparing the simulation steps' influences on the results. Ref. [100] studies optimal gliding reentry flight trajectory based on physical planning method. Firstly it introduces the mathematical model of multiple objectives optimization problems achieved by physical planning method, then brings the reentry optimization trajectory problem into the frame of physical planning to obtain the solution for the four objectives: maximizing the range considered, minimizing the thermal protection system, minimizing the peak value of heating-flow density, and the most stable trajectory. Taking the cone-shape reentry vehicle with wing as an example, the specific preference structure is determined by calculating and analyzing the single objective optimization result, the multiple objectives optimal trajectory is solved by genetic algorithm in consideration of heat flow, overload, dynamic pressure and terminal conditions constraints.

Besides, rush-bounce flight is a typical flight mode of hypersonic vehicle, which brings about many potential advantages to integrated performance and design technology of the vehicle. To minimize weight and maximize flight distance, by the way of analyzing the stress state and flight process of the vehicle, Yao [101] established an rush-jump flight trajectory optimization method based on the combination of the principle of function with calculus concepts as foundation, and then did flight numerical simulation. Taking successive bouncing flight of hypersonic vehicle in near space as research objective, Zhao et al. [102] set longitudinal plane motion model and flight trajectory optimization model. In consideration of the value and change rate of the attack angle, normal overload, dynamic pressure, heat flow and landing conditions constraints, the bouncing flight trajectory was optimized in a flying period by genetic algorithm, and flight trajectory was provided with minimum fuel consumption rate.

## 5 A summarization on cooperation planning of trajectory optimization for multiple flight vehicles

Cooperation planning of numerous flight vehicles is a complex multiple objectives optimization and decision problem with many constraints and strong coupling, this cooperation planning can effectively improve campaign performance. The total performance index of cooperation planning is to create flight trajectories for various flight vehicles and guarantee that the flight vehicles can fulfill the assigned flying task in certain period of time and reach the respective target points, try to avoid the threatening area. Such a flight path may not be the optimal trajectory to every single flight vehicle, but it must be a global optimal or suboptimal trajectory to the whole flight vehicle formation [103].

In conclusion, the main developed performance indexes of cooperative combat system for numerous flight vehicles can be basically described as follows: (1) To improve penetration capabilities of numerous flight vehicles; (2) To improve interception capabilities of numerous flight vehicles; (3) To improve electric counterplot capabilities of numerous flight vehicles; (4) To improve capabilities of tracking and hunting, as well as improve tracking accuracy of numerous flight vehicles; (5) To reduce the launching number of numerous flight vehicles; (6) To improve comprehensive fighting efficiency of numerous flight vehicles.

The main performance indexes of cooperative combat above includes the main operation modes of future flight vehicle weapon systems, which can reduce the combat cost, improve the liability of tasks fulfillment, enhance the quick-reaction capability and comprehensive fighting efficiency of the weapons together with target tracking, information sharing, intelligent decision and soft killing confrontation.

The algorithm of global optimization design on numer-

ous flight vehicles trajectory integration consists of two major steps specifically: ① optimal global organizational task allocation of attack-defense confrontation flight vehicle; ② an integrative global optimization on trajectory of attack-defense confrontation flight vehicle.

### 5.1 Organizational task allocation

As for  $n_A$  offensive flight vehicles with  $n_T$  targets, all possible remaining flight time  $t_{go\_ij}$  ( $i$  is the target number,  $j$  is the number of offensive flight vehicles,  $i \leq n_T$ ,  $j \leq n_A$ ) is as follows. This remaining flight time  $t_{go\_ij}$  includes all combinations of every offensive flight vehicles and every target (static or maneuvering) to be tracked.

$$T_{go} = \begin{bmatrix} t_{go\_11} & t_{go\_21} & \cdots & t_{go\_n_T 1} \\ t_{go\_12} & t_{go\_22} & \cdots & t_{go\_n_T 2} \\ \cdots & \cdots & \cdots & \cdots \\ t_{go\_1 n_A} & t_{go\_2 n_A} & \cdots & t_{go\_n_T n_A} \end{bmatrix} \quad (39)$$

The total value of all remaining flight time  $t_{go\_ij}$  for offensive flight vehicles is minimized:

$$T_{go}^* = \min_{1 \leq j_1 \leq n_A, 1 \leq k \leq n_T} \sum_{j_1=1}^{n_A} \sum_{k=1}^{n_T} t_{go\_j_1 k}^* \Rightarrow i^* = t_{go\_ij, \min}(i^*, j). \quad (40)$$

Get the attack division of each offensive flight vehicle  $j$  corresponding to (static or maneuvering) target flight vehicle  $i^*$  according to the results of minimum total value of remaining flight time. If a target is not inside the attack zones of the offensive flight vehicle  $j$  when calculating the remaining flight time, this remaining flight time  $t_{go\_ij}$  is invalid.

### 5.2 The algorithm of global trajectory optimization on many flight vehicles

With each party being in a complex environment (the future is unknown to both parties), the whole calculating steps are as follows for global integration optimal systematic fight simulation of group attack and defense confrontation among numerous flight vehicles.

(1) Both the offensive and defensive parties (flight vehicle of each party) conduct the integrative flight path planning on the basis of the combinational optimization algorithm, then send all of the command information about flight path planning to each (flight vehicle) unit. This process will be fulfilled on the decision system computers of the offensive and defensive parties, based on the information of all flight vehicle states. The process is as follows: a) The tasks allocation of all the flight vehicles of attacking and penetrating party; corresponding flight path planning of each flight vehicle of the offensive party and the attack order aiming at ground targets. b) The tasks allocation of all

the flight vehicles of defensive party; corresponding flight path planning of each flight vehicle of the defensive party and attack order aiming at the penetration party and interception order aiming at ground defensive system.

(2) A simulation on the flights and confrontation system fight is made on the basis of all of the information about flying conditions and environment, offense and defense strategies and the trajectories designed of both parties.

Regarding to the algorithms of integrative trajectory panning of numerous flight vehicles, it can be shown that most of the relevant problems about multiple trajectories cooperation planning belong to civilian and commercial domains according to some public references about cooperation planning of multiple trajectories [104–110], such as the collaborative unknown terrain detection of multiple flight vehicles, resource supply from the formation of multiple flight vehicles. However, the method of multiple trajectories cooperation planning in military and striking field is rarely mentioned. There is only a little research on multiple trajectories cooperation planning in China. And two kinds of methods are often used to study the flight trajectory optimal planning, one is the planning method designed on the basis of evolutionary algorithm in artificial intelligence field and the other one is the planning method based on visualized path searching. Of which, the former one is suitable for theoretical analysis, but it can hardly represent various information of planning constraints which make convergence rate hard to control in practice. While the latter one can easily integrate all information of various planning constraints, but it confronts problems such as combination explosion. To some extent, combination explosion problem can be well handled by adopting the trajectory planning method based on path searching and at the same time designing various conditions to reduce the search space.

In order to realize the highly coordinated attack of multiple anti-ship missiles in a constant wind, Pan [111] proposed an algorithm of cooperative path planning for multiple missiles in the presence of a known constant wind based on genetic algorithms. Firstly, the effect of a constant wind on the ground velocity and the flight time of a path of missile was analyzed. Secondly, how to evaluate paths was discussed in detail where asynchronism evaluation strategy changed along with the evolution of populations was adopted, and the operators of genetic algorithms were designed. Finally, simulation results indicated that the cooperative path planning algorithm could produce highly cooperative paths, which evaded threats successfully in the presence of a known constant wind. Ma et al. [112], taking the new concepts campaign modes such as Leader/attacking shell as example, discussed the integrated framework of cooperative combat of multiple flight vehicles on the basis of C<sup>4</sup>ISR system. They analyzed information fusion system, situation assessment system, task planning system, war damage evaluation system and data link system, then compared main algorithms of tasks planning such as Voronoi

map method, and finally designed the frame of hardware system and software module. To solve the coordination of rendezvous and collision avoidance of multiple aeronautical vehicles, Mao [113] proposed a cooperative trajectory planning method based on three-dimensional environment. This method introduces a new planning frame, which takes the coordination problem into account before and during the trajectory searching process. During the searching process, the method gives considerations to both the original constraint of every single trajectory and the collision check between multiple trajectories. Meanwhile, a heuristic search is also conducted by factors such as the uncoordinated value of the flight path. Li [114] proposed a path planning strategy for coordinated attack of multiple unmanned aerial vehicles (UA-Vs). An initial entrance flight path can be built by the Voronoi diagram according to the location of the known threats such as flight vehicles and radars. And B spline curves are used to modify the initial entrance flight path thus to create a feasible flight path for unmanned aerial vehicle. Coordinated attack paths for several unmanned aerial vehicles are obtained according to the conditions of the coordinated attack. Finally the exit trajectory of unmanned aerial vehicles from the target to the start position is researched, and the simulation is processed combining with specific problem. Zhang [115] proposed a path planning strategy for the coordinated multiple unmanned vehicles (UAVs). Firstly, an initial path from the projection start position to the target position can be devised by constructing a Voronoi diagram based upon the locations of the threats. Then the initial path is discretized for adjustment and smoothing. At last the desirable paths to the target are obtained by simulating the dynamic chain in a potential field. The result for multiple vehicles is a set of smooth and flyable paths with equal length that reduces exposure to threats. Based on the characteristics that the multiple selective paths can be planned for cooperative combat cruising missiles whose communication is supported by satellite data chain, Zhang put forward the path planning method for selective paths based on the algorithm of multi-path label correcting. The node taboo expanding strategy was introduced on the basis of traditional label correction algorithm in order to achieve the turning control and track interval control, guarantee the planning paths and satisfy the flight constraints of the flight vehicles. The 3D programming capability and threat-avoiding capability of this planning method were achieved by introducing the elevation information and threat information of digital terrain into weighted graph. And the designed steps of the selective flight path were also provide [116]. According to Xiao et al. [117], cooperative path planning and mission assignment should be considered at the same time in coordinated task planning, through which targets are allotted in reason, the attack flight paths are determined and all missiles arrive at their respective targets simultaneously. For these considerations, the optimized model is given for tasks assignment and

cooperative paths choosing and a coding structure is also provided. To overcome the shortages of genetic algorithm, some improvements are also put forward. Then simulation is performed on cooperative missions planning model by adopting the improved genetic algorithm. Ma [103] studied the cooperative path planning and target allocation approach based on battlefield environment with threats and no-fly zones was developed. Firstly, a Voronoi diagram considering battlefield constraint was completed, and the improved Dijkstra's algorithm was selected for obtaining the minimum-cost paths. Then the initial paths were processed dynamically by using LOS based path shortening method and path smoothing algorithm, and the flyable paths that could save fuel and satisfy missile' mobile performance were obtained. At last, taking the problem as a Multi-dimensional, Multi-choice Knapsack Problem (MMKP), the cooperative mission of multi-missile attacking multi-target was realized, and the multi-missile cooperative paths with a minimum overall mission cost were obtained.

In the study of group attack and defense confrontation, the global integrative optimization should be adopted by both parties of the group attack and defense confrontation. That is to say, when the offense party adopts the global integration optimal strategy (control of trajectory and each subsystem), the defense party also will adopt the global integrative optimization in consideration of the whole trajectory and control. Besides, both the attack and defense parties should take multidimensional unknown random disturbance factors into consideration.

## 6 Conclusions

Firstly, this paper briefly explains the fundamental principles of various kinds of commonly-used trajectory optimizations, their characteristics and applications; then introduces several frequently-used intelligent algorithms and combinational algorithm appearing in recent years; finally reviews the trajectory optimizations of multiple flight vehicle cooperation planning and hypersonic vehicle trajectory optimization.

In future, the research on trajectory optimization should be based on the background of calculation flight dynamics which should be greatly developed for the future flight vehicle systems, control theory and engineering research, so as to achieve the following functions of numerical optimization algorithm for optimal flight trajectories:

(1) Global optimization of multiple objective functions on the flight trajectory, i.e. making all of the possible performance index optimal under the conditions of meeting all the constraints.

(2) The global integrative optimization for group confrontation flight trajectories of multiple flight vehicles. This attack and defense confrontation optimization design refers to that at the same time both the attack and defense parties

adopt the optimal multidimensional strategy for the global perspective of integration.

(3) Real-time and on-line characteristic: the global integrative optimization of flight trajectory of multiple flight vehicles can be achieved in several seconds or milliseconds and be put into real-time use in the computers equipped inside the flight vehicles.

(4) High precision: it can provide the results of the optimized numerical calculation of flight trajectory at 7 degrees of freedom and complete precise information about each relevant subsystem, and meet the requirements of precision of the corresponding system.

(5) Be able to consider the effects of all various unknown random disturbances in different time and locations.

(6) A numerical algorithm of flight trajectory optimization should have all of the above functions.

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